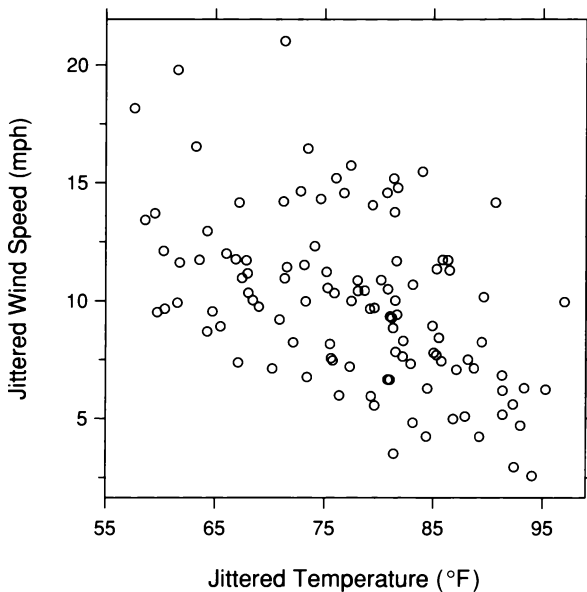


3.10 Bivariate Distributions

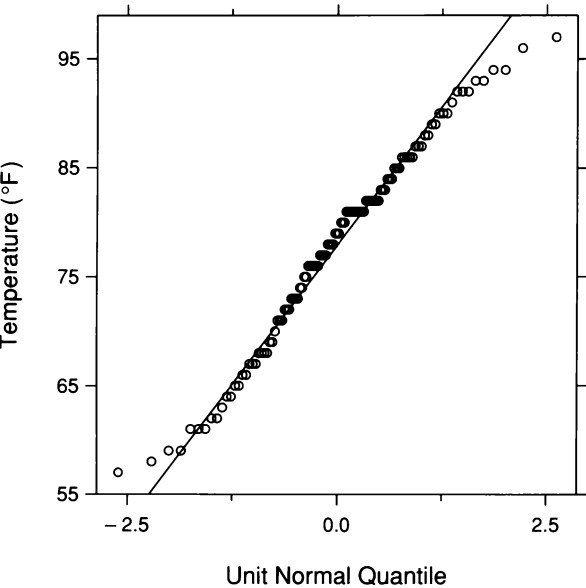
For the examples encountered so far in this chapter, the bivariate data consist of a response variable and a factor variable. In other words, the goal of each analysis has been to determine how one variable depends on the other. For many bivariate data sets, however, the goal is simply to determine the *bivariate distribution* of the data in the two-dimensional space of the measurements, rather than determining how the variation in one variable explains the variation in the other. Figure 3.56 shows an example. The bivariate data are 111 measurements of wind speed and temperature on 111 days from May to September of 1973 in the New York City metropolitan region [13]. The temperature measurement is the daily maximum. The wind speed measurement is the average of values at 0700 and 1000. The goal is not to determine how the variation in one explains the variation in the other, but rather to see how wind speed and temperature vary together.



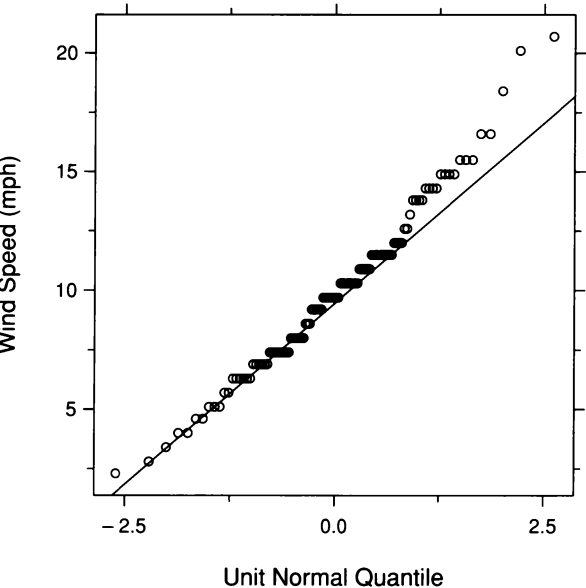
3.56 A scatterplot displays the wind speed and temperature data.

One aspect of a bivariate distribution is the two univariate distributions of the two separate sets of measurements. Figures 3.57 and 3.58 are normal q-q plots of the temperatures and wind speeds. The temperatures are slightly platykurtic. The wind speeds are slightly

skewed toward large values; one result of the univariate skewness for the bivariate distribution of the data is several stragglers at the top of the scatterplot of Figure 3.56.



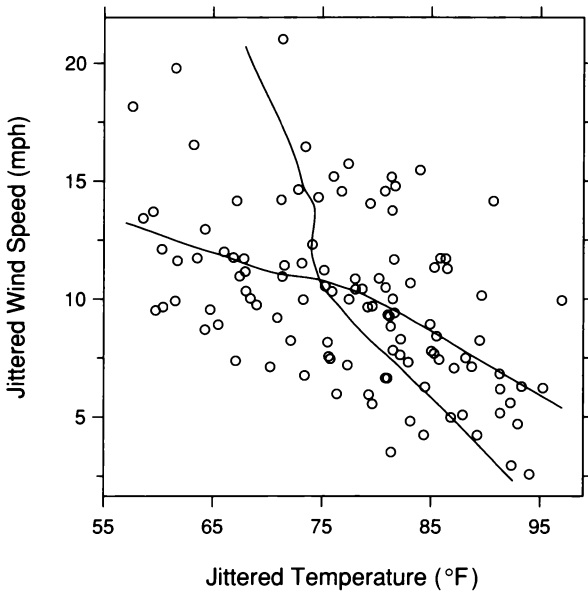
3.57 The univariate distribution of the temperatures is compared with the normal distribution by a normal q-q plot.



3.58 The univariate distribution of the wind speeds is compared with the normal distribution by a normal q-q plot.

Because temperature and wind speed are not designated as factor and response, it is no more informative to see a loess fit to wind speed as a function of temperature than it is to see a loess fit to temperature as a function of wind speed. For this reason, Figure 3.59 graphs both loess fits. The curve with the lesser overall slope is the fit to wind speed, and the curve with the greater overall slope is the fit to temperature.

In Figure 3.59 the fit to wind speed is nearly linear, but the fit to temperature has more curvature. The visualization shows that wind speed and temperature vary together; a decrease in wind speed tends to be associated with an increase in temperature.

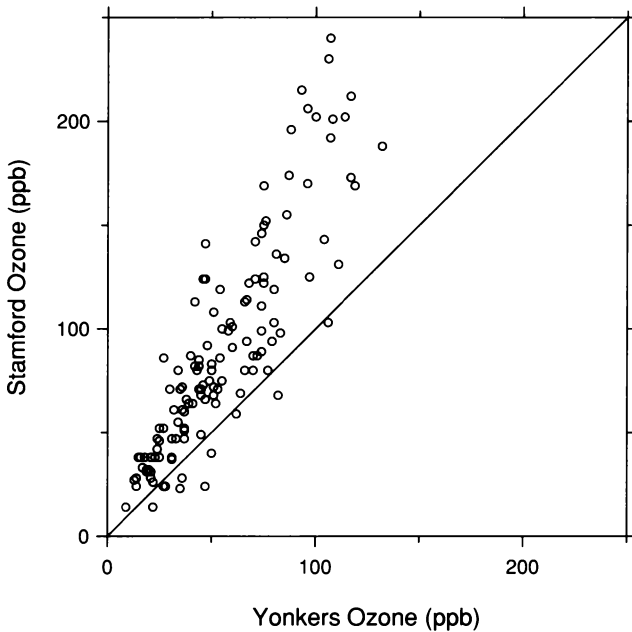


3.59 Two loess curves have been added to the scatterplot of wind speed and temperature. The loess parameters are $\alpha = 3/4$ and $\lambda = 1$.

Bivariate Data with the Same Units

It is common to encounter bivariate distributions where both variables have the same units. Most often, this arises because experimental units, or samples, are measured twice, for example, in two different locations or at two times or by two different measurement techniques. Figure 3.60 shows an example. The data are daily maximum ozone concentrations at ground level on 132 days from May 1, 1974 to September 30, 1974 at two sites in the U.S.A. — Yonkers, New York and

Stamford, Connecticut — which are approximately 30 km from one another [16]. The sample for each measurement is the air mass on a particular day, and the bivariate data arise from two measurements at the two sites.

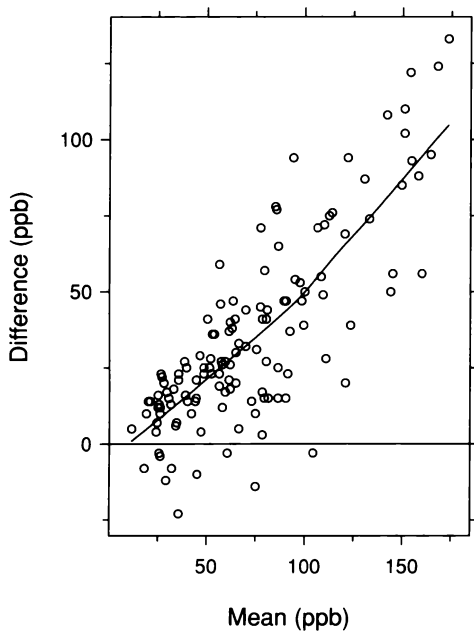


3.60 The scatterplot displays measurements of ground-level ozone concentration at two sites.

Tukey Mean-Difference Plots

Whenever the two axes of a graph of two variables have the same units, a Tukey mean-difference plot, or m-d plot, is a candidate for visualization. For example, m-d plots served as an adjunct display for q-q plots in Chapter 2. Figure 3.61 is an m-d plot of the points of Figure 3.60; the difference in the concentrations at the two sites on each day is graphed against the mean for the two sites. A loess curve has been added to aid our judgment of how the differences change with the mean level.

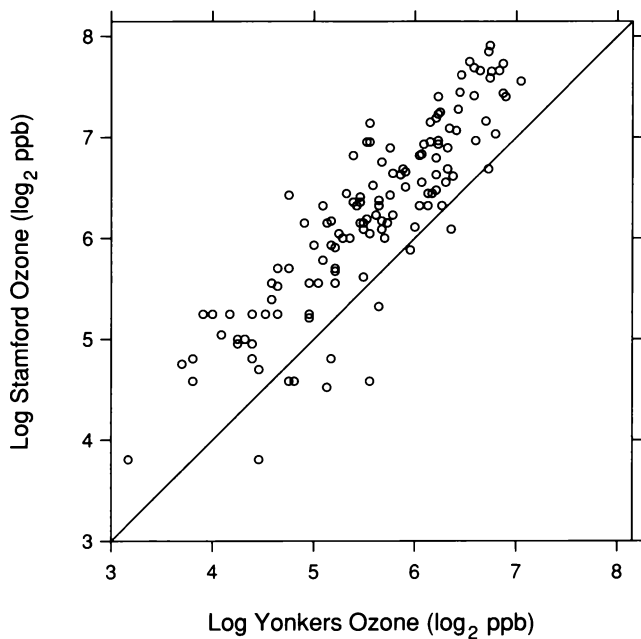
Figures 3.60 and 3.61 show that ozone concentrations at the two sites are correlated. This is to be expected since both are subject to the same weather systems, and the variation in these systems account in large part for the variation in the concentrations through time. It is also clear that



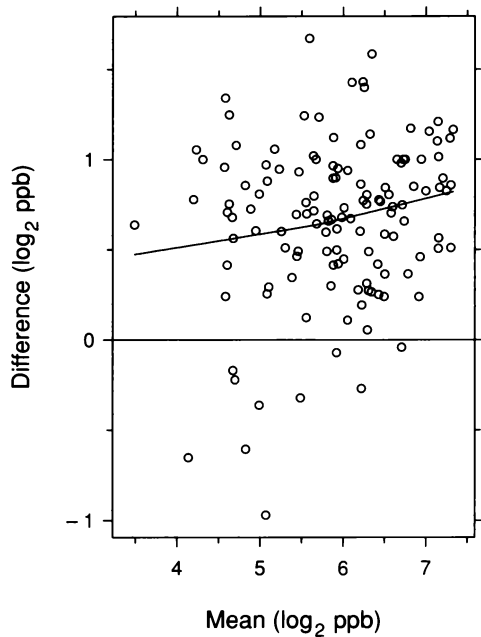
3.61 The Tukey mean-difference plot graphs ozone differences against ozone means. The parameters of the loess curve are $\alpha = 1$ and $\lambda = 1$. The curve is banked to 45° .

concentrations at Stamford are higher than those at Yonkers; on all but 10 days, Stamford values exceed Yonkers values. Stamford tends to have higher concentrations because it lies downwind of the New York City metropolitan region; the chemical reactions that produce ozone are able to more effectively create the air pollutant once the air mass moves away from the heaviest concentrations of primary pollutants. The amount by which Stamford exceeds Yonkers tends to increase as the levels at the two sites increase. The loess curve in Figure 3.61 shows that the differences increase from about 0 ppb on the cleanest days to about 100 ppb on the dirtiest days.

The m-d plot shows the concentration differences have a complex behavior. But it may be that concentration ratios have a simpler behavior. We can check this by taking logs and making the scatterplot and m-d plot again. This is done in Figures 3.62 and 3.63. The loess curve still increases showing there is an increase in the percentage difference between Stamford and Yonkers as the overall level of ozone increases. But the effect on the log scale is less dramatic. At low concentrations, Stamford exceeds Yonkers by $0.5 \log_2$ ppb, which is a factor of $2^{0.5} = 1.4$. At high concentrations, the difference is $0.9 \log_2$ ppb, which is a factor of $2^{0.9} = 1.9$.



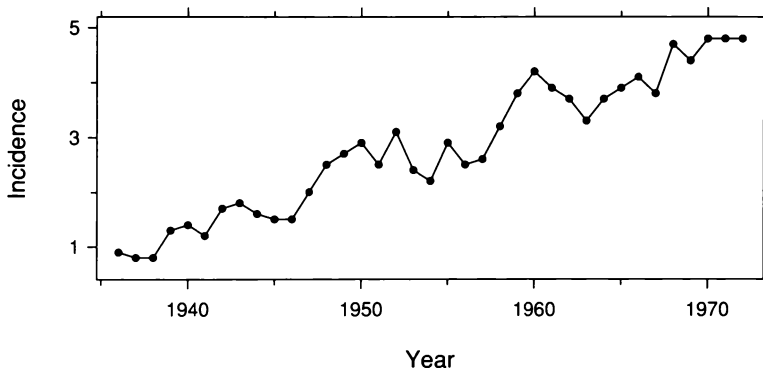
3.62 The scatterplot displays the logs of the ozone measurements.



3.63 The m-d plot graphs log ozone differences against log ozone means. The parameters of the loess curve on the plot are $\alpha = 1$ and $\lambda = 1$. The curve is banked by an aspect ratio of 1.5. Banking to 45° results in an aspect ratio that is too large.

3.11 Time Series

Figure 3.64 displays yearly incidences of melanoma, the deadly form of skin cancer [53]. The units are age-adjusted cases per 10^6 population. The data are from the state of Connecticut in the U.S.A. A principal cause of melanoma is exposure to solar radiation. For white populations, incidence tends to increase as latitude decreases because of increased exposure to radiation, and incidence is higher at skin sites with more exposure to the sun, such as the face.



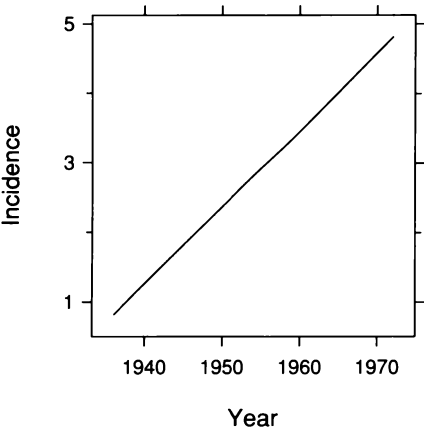
3.64 A time series — yearly age-adjusted melanoma cases per 10^6 population in Connecticut, U.S.A. — is graphed with 45° banking of the local segments, the line segments connecting successive observations.

The melanoma data are a time series: measurements of a quantitative variable through time. Melanoma incidence is the response and time is the factor; the goal is to see how incidence varies through time. While the data type is still bivariate data, the special character of the time variable changes the strategy and methods of analysis.

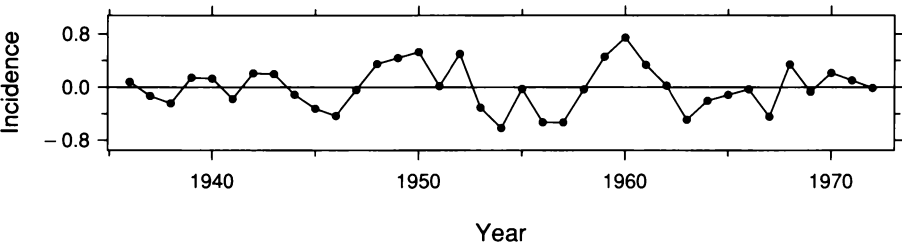
Time Components

Figure 3.64 shows a steady upward trend in the data. Superposed on this trend are oscillations that appear to rise and fall with periods of about 10 years. Figure 3.65 graphs a loess curve fitted to the data to describe the upward trend; the rise is very nearly linear. Figure 3.66 graphs the residuals from the loess fit against time. The oscillations are contained in the residuals since they are not fitted by the trend curve.

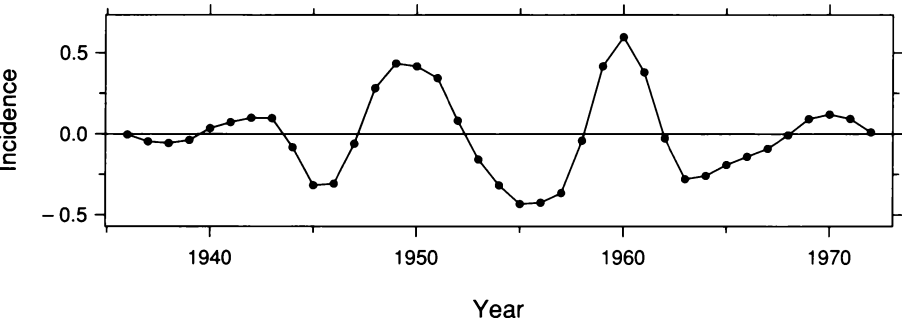
Figure 3.67 graphs a loess curve that was fitted to the residuals to describe the oscillations. There are approximately three complete oscillations; since the data cover 36 years, the average period is about 12 years.



3.65 The display graphs a loess trend fit to melanoma incidence. The local segments are banked to 45°.



3.66 The display graphs the residuals from the loess trend fit to melanoma incidence. The local segments are banked to 45°.



3.67 The display graphs an oscillatory component fitted by loess to the residuals from the trend fit. The local segments are banked to 45°.

This initial visualization of the melanoma data illustrates the principal consequence of having time as a factor. Instead of fitting a single curve to describe dependence — as we did for the previous bivariate data sets in the chapter — we often fit multiple curves to the data to describe multiple *time components*. For the melanoma data, one component is the trend component and another is the oscillatory component. Separate descriptions of different time components of a time series are necessary because different components often have different causes.

Loess Fitting for Time Series

Loess will be our principal tool for fitting time components, which are often complicated functions that defy description by simple parametric functions. We will suppose that the time series under analysis has been measured at equally spaced points in time; this is the case for many series in practice.

Consider the method for computing the loess fit, $\hat{g}(x)$, at a time point x . The first step is to multiply the number of observations, n , times the smoothing parameter, α , and then truncate the result to an integer, q . Then we find the q observation times closest to x and carry out the local fitting using data at these times. For time series applications, we will always take α to be of the form ℓ/n , where ℓ is an integer; thus $q = \ell$, and we will quote ℓ rather than α because it is typically more natural to think in terms of ℓ for a time series. For the trend fit to the melanoma data, $\ell = 30$ and $\lambda = 1$. For the fit to the residuals from the trend fit, $\ell = 9$ and $\lambda = 2$. Locally linear fitting is used for the trend fit because the trend has little or no curvature. Locally quadratic fitting is used for the oscillatory fit because the oscillations have peaks and valleys. The values of ℓ arose from experimenting with several values.

For a times series, we will typically evaluate a loess fit at the times at which the series is measured. This was done to display the loess fits in Figures 3.65 and 3.67.

Repeated Loess Fitting

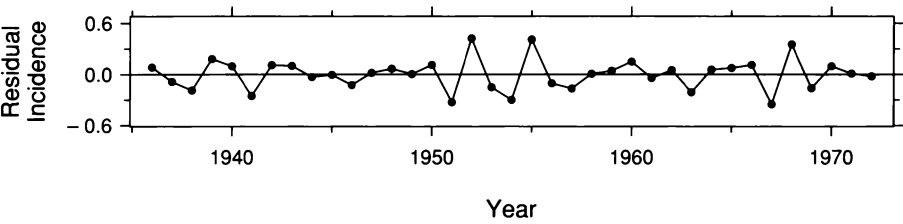
The above fitting of two curves to the melanoma data is just the first step, or pass, of *repeated loess fitting*, a method for fitting two or more time components. For the first pass, the component with the largest value of ℓ is fitted by loess. For the melanoma data, this is the trend component graphed in Figure 3.65. Then the component is subtracted from the data to form residuals. The component with the second largest ℓ is fitted to the residuals. For the melanoma data, this is the oscillatory component graphed in Figure 3.67. If there are more than two time components to be fitted, new residuals are computed by subtracting the two fitted components, the next component is fitted, and so forth.

In the next pass, each component is fitted again by subtracting all other fitted components and fitting a loess curve. For example, in a second pass through the melanoma data, we subtract the oscillatory component from the first pass and fit the trend component to the residuals. Then we subtract this new trend component from the data and fit the oscillatory component to the residuals.

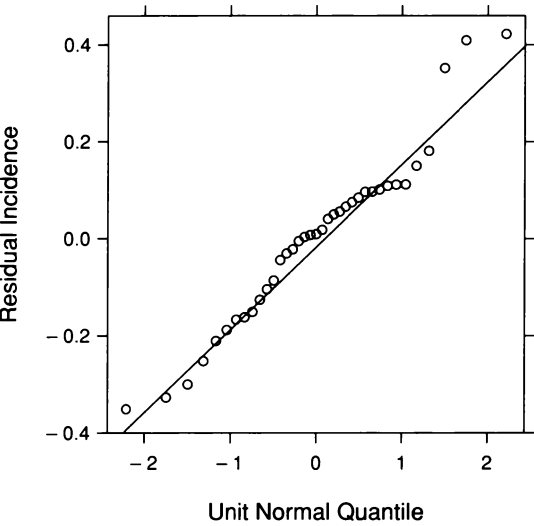
Several passes can be made through all components until the results stabilize, which typically happens after a small number of passes. The reason for multiple passes is that in some cases, different components can compete for the same variation in the data. The potential for competition increases as any two values of ℓ get closer. The multiple passes are needed for the components to settle down and decide which component gets what variation. For the melanoma data, additional passes beyond the first produce fits nearly identical to those from the first pass, so there is virtually no competition at all, and Figures 3.65 and 3.67 show what amount to the final fits.

Visualizing the Final Residuals

Figure 3.68 graphs the residuals from the two components of the fitting to the melanoma data. No major effect appears. One minor peculiarity is that the three largest residuals are preceded by the three smallest residuals. Figure 3.69 is a normal q-q plot of the residuals. The distribution of the data appears well approximated by the normal. For example, there are no outliers or more pervasive leptokurtosis. This is comforting because the loess fitting did not employ the bisquare robustness feature. Were such robustness needed, we would compute bisquare weights at the end of each pass and use them in each fitting of the next pass.

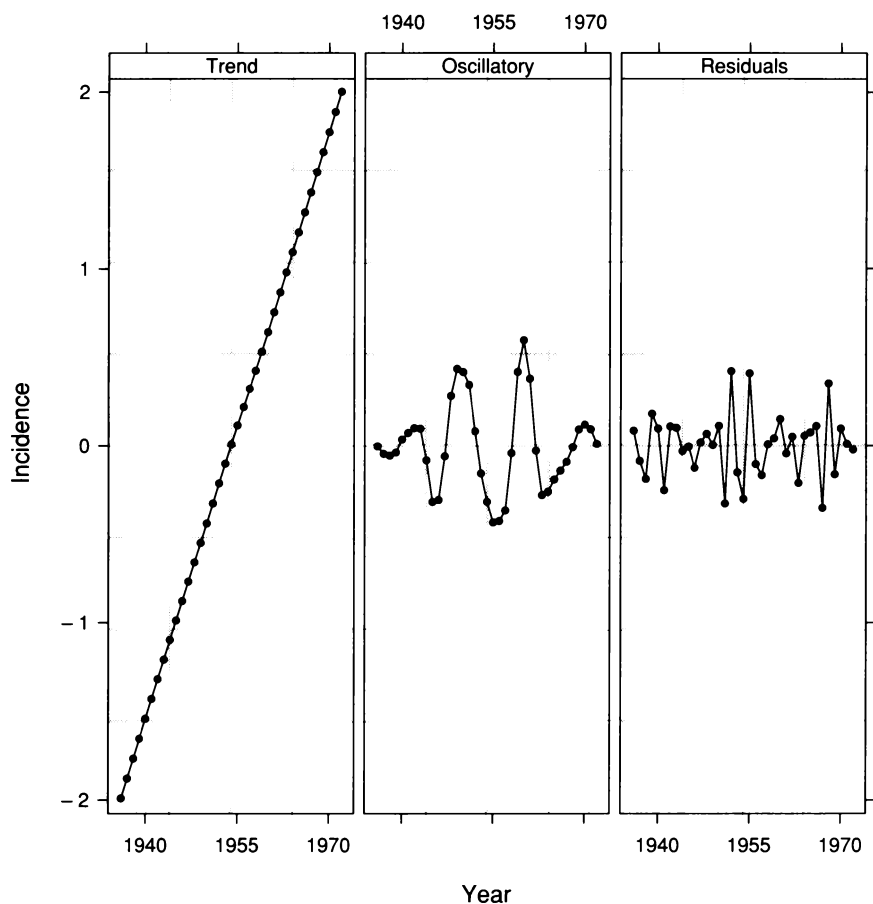


3.68 The display graphs the residuals from the fitting of two time components to melanoma incidence. The local segments are banked to 45°.



3.69 A normal q-q plot displays the incidence residuals.

For the visualizations of the data earlier in the chapter, r-f spread plots showed the relative amounts of variation in fitted values and residuals. For a time series fitting with multiple components, we want to compare the amount of variation not only in the overall fit and the residuals, but also in the different components as well. This can be done simply by graphing the components and the residuals against time as shown in the r-f spread plot of Figure 3.70 for the melanoma fits. The trend panel shows the yearly trend values minus their mean; the resulting values are centered on zero as are the values of the other two components. The trend component accounts for a large amount of the variation in the data. The oscillatory has less variation, and the residual component has the least.

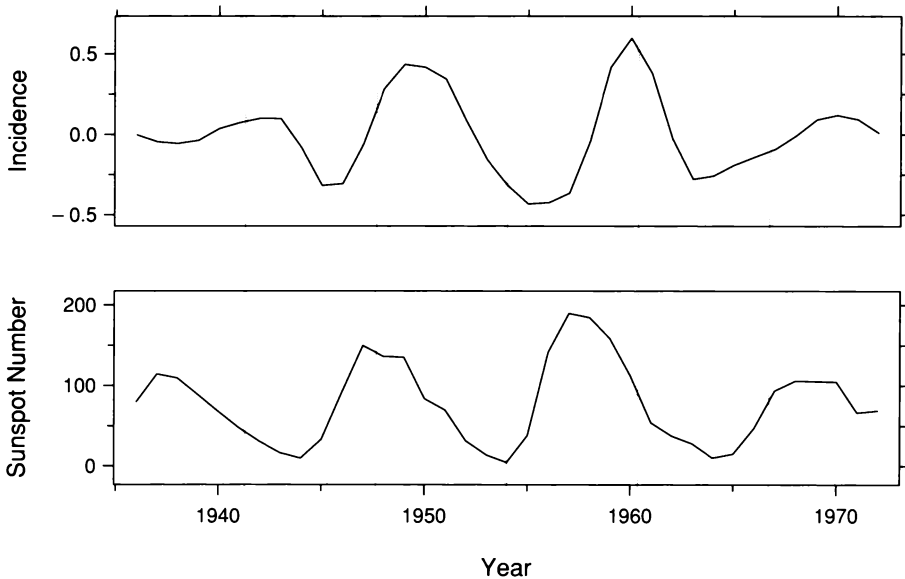


3.70 An r-f spread plot compares the variation in the three time components of melanoma — the trend component, the oscillatory component, and the residuals.

Causal Factors for the Melanoma Data

The trend component in the melanoma data has a number of possible causes. One, of course, is simply more exposure — for example, people spending more time at the beach. A second is more widespread and accurate diagnosis. A third is a population with greater susceptibility. No firm explanation of the cause is known at this time.

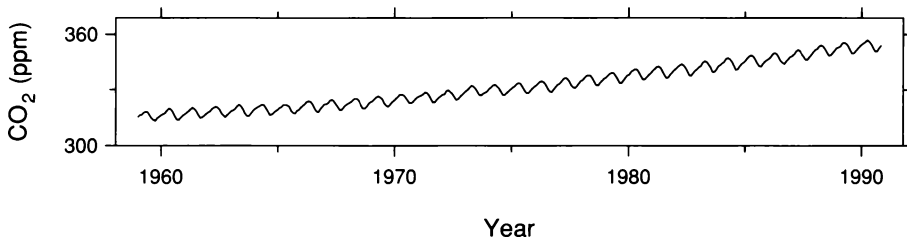
The oscillatory component has a firm explanation. The cycles have periods of about 12 years. Given that solar radiation is a cause of melanoma, given that the sunspots have an 11-year cycle, and given that solar activity increases with increasing sunspot number, it seems quite likely that the oscillatory component and the sunspot numbers are related. Figure 3.71 graphs both the oscillatory component and the sunspot numbers on juxtaposed panels. There is an obvious association between the two series in which the peaks and troughs of the sunspots occur somewhat earlier. This time delay, or lag, is not surprising. An increase in radiation would not necessarily trigger melanoma cases instantaneously but rather would trigger them at some point in the future. And diagnosis once symptoms appeared would not occur immediately.



3.71 The oscillatory component fitted to melanoma incidence is compared with the sunspot numbers.

3.12 Seasonal Components in Time Series

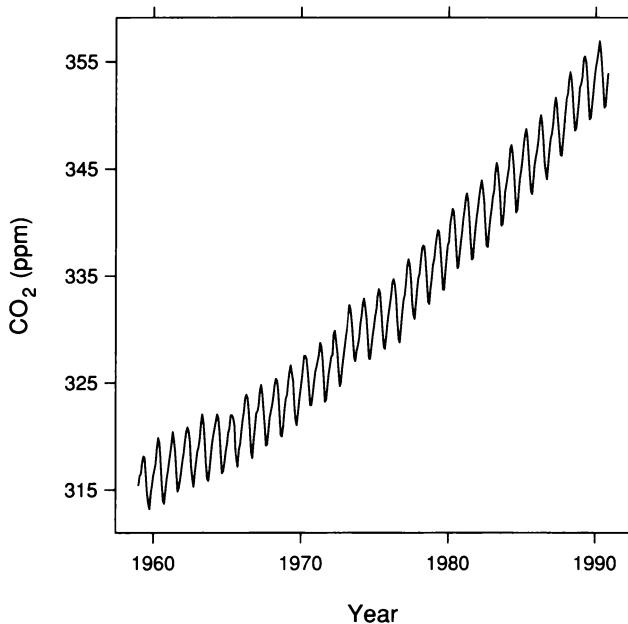
Figure 3.72 displays monthly average CO₂ concentrations at the Mauna Loa observatory in Hawaii [10, 58]. One time component revealed by the graph is a persistent upward trend. This is the result of man's emissions of CO₂ into the atmosphere. The rise, if continued unabated, will cause atmospheric temperatures to rise, the polar ice caps to melt, the coastal areas of the continents to flood, and the climates of different regions of the earth to change radically [70].



3.72 The display shows a time series, monthly CO₂ concentrations at Mauna Loa. The local segments are banked to 45°.

The aspect ratio in Figure 3.72 has been chosen by 45° banking of the local segments, the line segments connecting successive monthly observations. This gives an opportunity to study local behavior, but it does not serve our assessment of the trend particularly well. If the segments of a trend curve were banked to 45°, the aspect ratio of the graph would be close to one.

Figure 3.73 graphs the data again with an aspect ratio of one to provide a better visualization of the trend component. The graph suggests there is convexity in the trend; the rate of increase of CO_2 is itself increasing through time.



3.73 The CO_2 time series is graphed again with an aspect ratio equal to 1 to approximately bank the underlying trend to 45° .

A second time component in the CO_2 data is oscillations with a period of one year. The cause is foliage in the Northern Hemisphere. When the amount of foliage begins to increase in the spring, CO_2 is absorbed from the atmosphere, and when the amount of foliage begins to decrease in the fall, CO_2 is returned to the atmosphere. This component has a special property. It operates like a good clock; it is a *seasonal component*. The driving mechanism, the change in foliage throughout the year in the Northern Hemisphere, is tied to the revolution of the earth around the sun. As time has marched along, peaks have occurred in late spring and troughs have occurred in the fall. Unless man's activities disrupt the mechanism, we can count on this behavior for the future, and we could accurately predict the occurrences of peaks in 100 years. Oscillations in a time series do not always keep good time. For example, the cycles in the sunspot series, a few of which are graphed in Figure 3.71, are not a particularly good clock. Over short time intervals of a few cycles, peaks can be predicted, but it is not possible to accurately predict peaks many decades into the future.

Figures 3.72 and 3.73 provide helpful visualizations of the CO₂ time components, but we can expect to see considerably more by fitting. We could proceed as we did with the melanoma data — fit the trend with a large value of ℓ and with $\lambda = 1$, and then fit the seasonal oscillations with a small value of ℓ and with $\lambda = 2$. But the excellent clock in the seasonal component gives an opportunity to take another tack that produces a better fit.

Seasonal Loess

Seasonal loess is a procedure for fitting a seasonal component. The procedure is iterative in the same way that repeated loess fitting is iterative, with several passes and with the fitting of components on each pass [18]. For seasonal loess, each pass consists of fitting two components. One is the seasonal component; seasonal loess has a special method for fitting this component. The other is a component whose sole purpose is to enable an uncontaminated fitting of the seasonal component; it captures long-period variations, which are trends and oscillations whose periods are longer than that of the seasonal period, which is 12 months for the CO₂ data. First, the seasonal component is fitted. Then the long-period component is fitted to the residuals. This is the first pass. In subsequent passes, each component is fitted to residuals from the other component. Usually, three passes are quite sufficient. If robustness is needed, bisquare weights can be computed at the end of each pass and used in each fitting of the next pass.

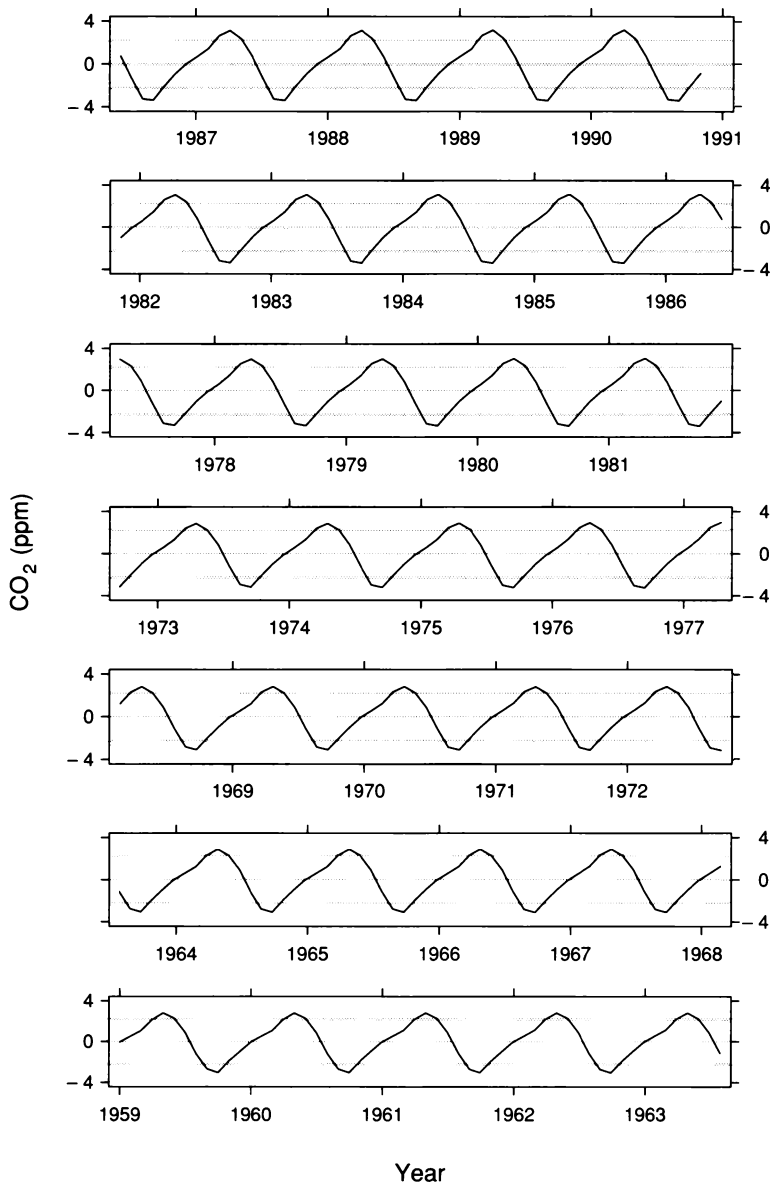
To fit the seasonal component, loess is applied separately to each *cycle subseries* of the residuals from the current long-period component. A cycle subseries consists of all data at one position of the cycle. For example, for the CO₂ data there are 12 subseries: the January values, the February values, and so forth. So to fit the seasonal component for CO₂ there are 12 separate loess smoothings, one for each subseries; the 12 fits are then combined to form the complete seasonal component. The details of the fitting for seasonal loess are given at the end of this section.

To carry out seasonal loess, values of ℓ and λ for the loess fits to the subseries must be chosen. For the CO₂ data, some experimentation showed that $\ell = 25$ and $\lambda = 1$ resulted in a seasonal component that was smooth and that adequately tracked the seasonal pattern in the data.

Cut-and-Stack Plots

The seasonal fit to the CO₂ data is displayed in Figure 3.74. A *cut-and-stack* display method is used to enable banking to 45°. Suppose we attempted to graph the CO₂ seasonal component on a single panel with 45° banking of the local segments. The resulting aspect ratio would be 0.017; and a single-panel graph with this aspect ratio that was 2 cm high would be 1.18 meters wide, too big for either a book page or a computer monitor. The solution is to take the single long panel, cut it into pieces, and stack the pieces up as in Figure 3.74. The seasonal component consists of 384 values, one value for each month. If the component were graphed on a single panel, the data would be displayed by 383 line segments connecting successive values. The cut-and-stack method also uses 383 segments because each cut occurs where one line segment joins the next one.

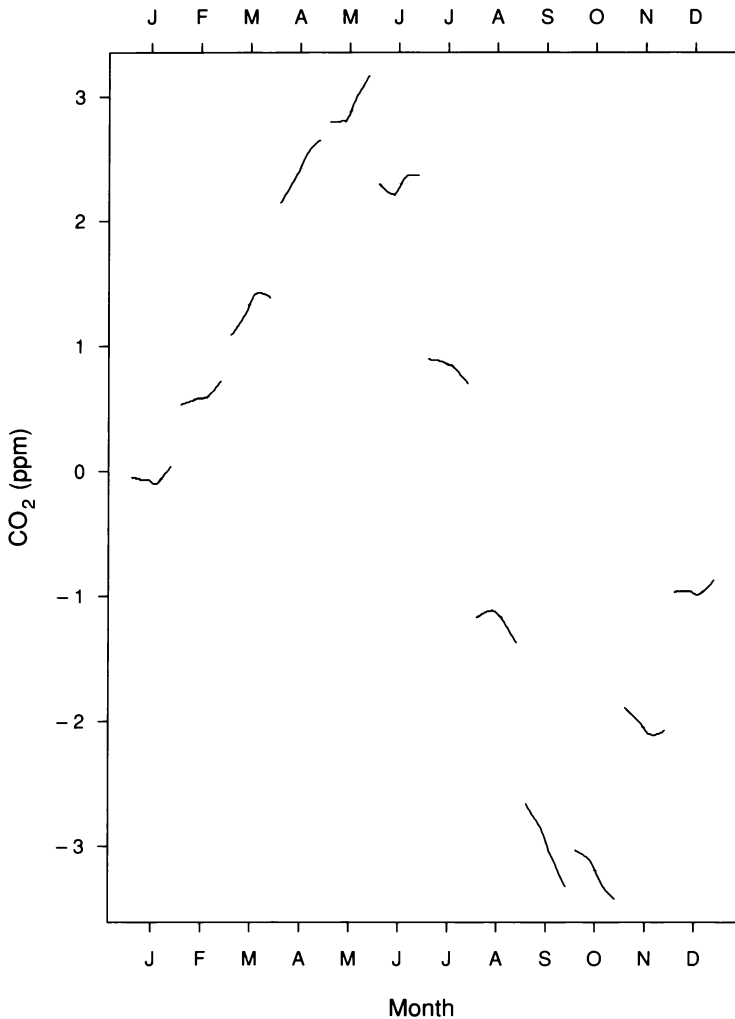
Figure 3.74 shows the seasonal cycles are all quite similar, but one change is apparent; the sizes, or amplitudes, of the seasonal oscillations increase slightly though time. The next visualization method provides a more sensitive look at this behavior.



3.74 The seasonal component fitted to the CO_2 data is cut and stacked so that the curve can be banked to 45° .

Cycle Plots

Figure 3.75 is a *cycle plot*, another method for graphing a seasonal component [27]. The cycle subseries are graphed separately. First, at the left of the data region, the January values are graphed for successive years, then the February values are graphed, and so forth. For each cycle subseries, the mean of the values is portrayed by a horizontal line segment. The graph shows the overall pattern of the seasonal cycles — for example, as portrayed by the horizontal mean lines — and also shows the behavior of each monthly subseries. Thus we can readily see



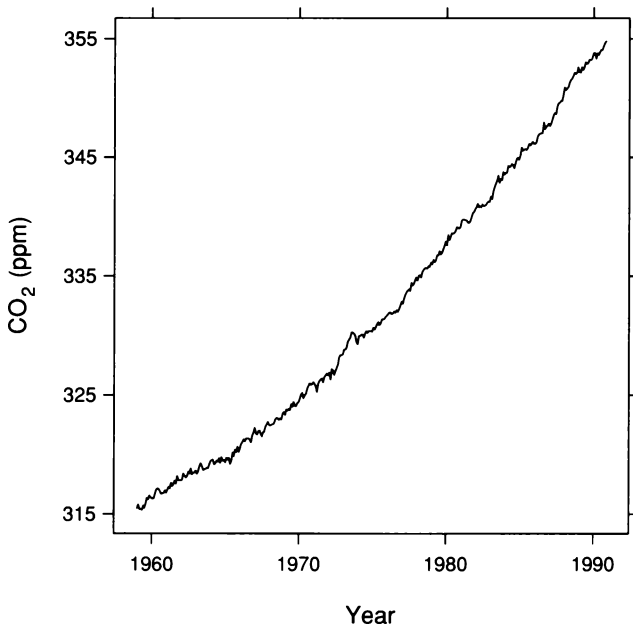
3.75 The monthly subseries of the CO_2 seasonal component are displayed by a cycle plot. The segments that make up the 12 curves are banked to 45° .

whether the change in any subseries is large or small compared with the overall pattern of the seasonal component.

Figure 3.75 shows interesting features. The first is the overall seasonal pattern, with a May maximum and an October minimum. The second feature is the patterns in the individual monthly subseries. For months near the yearly maximum, the subseries tend to be increasing rapidly although the overall increases are small compared with the seasonal amplitude; the biggest increases occur for April and May. Near the yearly minimum, cycle subseries tend to be decreasing; the biggest decreases are for September and October. The net effect is the increase in the amplitudes of the seasonal oscillations observed earlier. At present, no one knows if this has been caused by human activity or whether it is part of a natural cycle.

Fitting the Residuals from a Seasonal Component

Once a seasonal component has been fitted, other time components can be studied by fitting the residuals from the seasonal component. The CO₂ residuals, which still contain the trend component, are graphed in Figure 3.76.



3.76 The residuals from the seasonal component are displayed.